



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

EC3003-1 – DIGITAL SIGNAL PROCESSING
(E&C)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- The circular convolution of two sequences in time domain is equivalent to
 - Multiplication of DFTs of two sequences
 - Summation of DFTs of two sequences
 - Difference of DFTs of two sequences
 - Square of multiplication of DFTs of two sequences
- If the DFT of a sequence $x(n)$ is $X(k)$, then the DFT of $x(n)W_N^{-ln}$ is
 - $X((k+l))_N$
 - $X((k-l))_N$
 - $-X((k-l))_N$
 - None of these
- Parseval's theorem is defined by
 - $\sum_{n=0}^{N-1} |x(n)|^2 = \sum_{k=0}^{N-1} |X(k)|^2$
 - $\frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 = \sum_{k=0}^{N-1} |X(k)|^2$
 - $\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$
 - $\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N^2} \sum_{k=0}^{N-1} |X(k)|^2$
- If $x(n) = (1, 2, 0, 3, -2, 4, 7, 5)$, the DFT sample $X(0)$ is
 - 10
 - 25
 - 20
 - 30
- The overlap save method is used to calculate
 - The discrete convolution between a sampled signal and a finite impulse response (FIR) filter
 - The discrete convolution between a sampled signal and an infinite impulse response (IIR) filter
 - The discrete convolution between a very long sequence and an infinite impulse response (IIR) filter
 - The discrete convolution between a very long sequence and a finite impulse response (FIR) filter
- Overlap-Add Method Deals with principles that
 - The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $L + M - 1$
 - The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $L + M$
 - The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $2L + M - 1$
 - The linear convolution of a discrete-time signal of length L and a discrete-time signal of length M produces a discrete-time convolved result of length $2L + 2M - 1$
- The number of complex multiplications required for 500 point DFT computation using FFT (approximately) is
 - 500×499
 - 2293
 - 500×200
 - 4586
- Radix - 2 FFT algorithm performs the computation of DFT in
 - $(N/2) \log_2 N$ multiplications and $2 \log_2 N$ additions
 - $(N/2) \log_2 N$ multiplications and $N \log_2 N$ additions
 - $\log_2 N$ multiplications and $(N/2) \log_2 N$ additions
 - $N \log_2 N$ multiplications and $(N/2) \log_2 N$ additions

9. Identify the type of the filter that has the transfer function $|H(\Omega)|^2 = \frac{1}{1+(\Omega/\Omega_c)^{2N}}$
- A) Chebyshev Filters B) Butterworth Filters
C) Elliptic Filters D) None of the above
10. Task of obtaining a digital filter from the analog IIR filter involves developing a
- A) mapping from s-plane to z-plane B) mapping from z-plane to s-plane
C) Mapping from s-plane to s-plane D) mapping from z-plane to z-plane
11. In Chebyshev filter design approximation, the poles are plotted on the circumference of
- A) Circle B) Sphere
C) Hexagon D) Ellipse
12. The number of crests in passband ripples indicate the
- A) twice the order of the filter B) half the order of filter
C) order of filter D) not related to order of filter
13. Which of the following windows has a time domain sequence $h(n) = 0.5(1 - \cos \frac{2\pi n}{M-1})$?
- A) Bartlett window B) Blackman window
C) Hamming window D) Hanning window
14. Which of the following condition should the unit sample response of a FIR filter satisfy to have a linear phase?
- A) $h(n) = h(M-1-n)$; $n = 0, 1, 2 \dots M-1$ B) $h(n) = \pm h(M-1-n)$; $n = 0, 1, 2 \dots M-1$
C) $h(n) = -h(M-1-n)$; $n = 0, 1, 2 \dots M-1$ D) None of the mentioned
15. Which of the following gives the equation for group delay?
- A) $d\theta(\omega)/d\omega$ B) $\theta(\omega)$
C) $-d\theta(\omega)/d\omega$ D) $-\theta(\omega)$
16. What is the output of the single stage lattice filter if $x(n)$ is the input?
- A) $x(n) + Kx(n+1)$ B) $x(n) + Kx(n-1)$
C) $x(n) + Kx(n-1) + Kx(n+1)$ D) $Kx(n-1)$
17. The architecture that has two memories
- A) Von Neumann architecture B) VLIW architecture
C) Modified Harvard Architecture D) Harvard Architecture
18. The format is commonly used in Modern DSP processors
- A) Floating point B) Fixed point
C) IEEE floating point D) None of these
19. The basic arithmetic operations will not include one of these
- A) ADD B) SUBTRACT
C) MULTIPLY D) AND
20. Limiting the accumulator contents to its maximum limits is achieved by a simple logic circuit known as
- A) accumulation logic circuit B) stack logic circuit
C) saturation logic circuit D) none of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) For a length-8 sequence, $x(n) = (0, 1, 2, 3, 4, 5, 6, 7)$ find DFT $X(k)$ using DIT-FFT algorithm.	8	L*3	2	1
	b) A length-8 sequence $x(n) = (-4, 5, 2, -3, 0, -2, 3, 4)$ has an 8 point DFT as $X(k)$. Determine the sequence $y(n)$ whose 8-point DFT is given by $Y(k) = W_4^{3k} X(k)$ using appropriate property.	4	L3	1	1
	c) Let $x(n) = (1, 2, 0, 3, -2, 4, 7, 5)$. Evaluate the following. (i) $X(0)$ (ii) $X(4)$ (iii) $\sum_{k=0}^7 X(k)$ (iv) $\sum_{k=0}^7 X(k) ^2$	4	L3	1	1
2.	a) Given two sequences $x(n) = (1, 2, -1, 2, 3, -2, -3, -1, 1, 1, 2, -1)$ and $h(n) = (1, 2)$ with a block length of 4. Perform long sequence filtering by using a method that has aliasing effect.	8	L3	2	1
	b) Using DFT-IDFT method perform 4 point circular convolution of the two sequences $x(n) = (-3, 2, 4)$ and $h(n) = (2, -4, 0, 1)$.	4	L3	1	1

- c) Given $X(k)$ is a 14-point DFT of a real sequence $x(n)$. The first 8 samples of $X(k)$ are given by:
 $X(0) = 12$, $X(1) = -1 + j3$, $X(2) = 3 + j4$, $X(3) = 1 - j5$,
 $X(4) = -2 + j2$, $X(5) = 6 + j3$, $X(6) = -2 - j3$ and $X(7) = 10$.
 Find the remaining samples using appropriate property. Also evaluate

$$\sum_{n=0}^{13} e^{j\frac{4\pi n}{7}} x(n).$$

4 L3 1 1

3. a) Perform block convolution of the two sequences given $x(n) = (1, 2, 3, 4, 5, 6, 7)$ and $h(n) = (1, 2)$ with a block length of 4 using a method that avoids aliasing effect.
 b) Using relevant properties, find the DFT of the sequence, $x_c(n) = \cos \frac{2\pi}{N} k_0 n$, $0 \leq n \leq N - 1$.
 c) Compute IDFT of the sequence $X(k) = (0, 2, 0, 2)$ using DIF-FFT algorithm.

8 L3 2 1

4 L3 1 1

4 L3 2 1

Unit – II

4. a) Design a Butterworth low pass filter with PB attenuation as -1 dB and PB frequency as 4 rad/sec. SB attenuation is -20 dB with SB frequency as 8 rad/sec.
 b) Realize the direct form-I and direct form-II structure for the system function.

8 L3 3 1

$$H(z) = \frac{z^{-1} - 3z^{-2}}{10 + 4z^{-1} + 4.5z^{-2} - 0.5z^{-3}}$$

8 L3 3 1

5. a) The desired frequency response of an LPF is:

$$H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega}; & -\frac{3\pi}{4} \leq \omega \leq \frac{3\pi}{4} \\ 0; & \frac{3\pi}{4} < |\omega| \leq \pi \end{cases}$$

Design it using Hamming window ($M=7$) and obtain the system function.

8 L3 4 1

- b) An FIR filter is given by the difference equation as:

$$y(n) = x(n) + \frac{2}{5}x(n-1) + \frac{3}{4}x(n-2) + \frac{1}{3}x(n-3)$$

Realize this using lattice structure.

8 L3 4 1

6. a) The normalized transfer function of a second order Butterworth filter is given by:

$$H_2(s) = \frac{1}{s^2 + 1.414s + 1}$$

Convert the analog filter to a digital filter with cut-off frequency of 0.5π rad/sec using Bilinear transformation at $T=1$ sec.

8 L3 3 1

- b) Determine the impulse response of a FIR filter with reflection coefficients $K_1 = 0.6, K_2 = 0.3, K_3 = 0.5, K_4 = 0.9$ and $m = 4$. Also realize the direct form structure.

8 L3 4 1

Unit – III

7. a) Explain Arithmetic Logic Unit with a neat block diagram.
 b) Differentiate between fixed point and floating point processors.
 c) Find the signed Q15 representation for the decimal number -0.160123.
8. a) Describe Braun Multiplier for unsigned number multiplication. Highlight the issues observed in multiplication.
 b) Briefly discuss the concept of overflow and underflow in MAC operation.
 c) Convert the Q15 signed number 0.100011110110010 into its decimal equivalent.

8 L2 5 1

4 L2 5 1

4 L3 5 1

8 L2 5 1

4 L2 5 1

4 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
